Advanced Algebra

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Simplifying rational expressions**

Combine into a single fraction: $\frac{2}{x+1} + \frac{3}{x-1}$.

$\frac{2}{x+1} + \frac{3}{x-1} = \frac{2(x-1)+3(x+1)}{(x+1)(x-1)} = \frac{2x-2+3x+3}{x^2-1} = \frac{5x+1}{x^2-1}$.

2 **[Foundational]** • **Target: Polynomial division, remainder & factor theorems**

Use the remainder theorem to find the remainder when $f(x) = x^3 + 2x^2 - 5x + 1$ is divided by $(x - 2)$.

$f(2) = 8 + 8 - 10 + 1 = 7$. The remainder is 7.

3 **[Foundational]** • **Target: Polynomial division, remainder & factor theorems**

Given $f(x) = x^3 + 3x^2 - 4x - 12$:

a Use the factor theorem to show that $(x + 3)$ is a factor of $f(x)$.

$f(-3) = (-27) + 3(9) - 4(-3) - 12 = -27 + 27 + 12 - 12 = 0$, so by the factor theorem $(x + 3)$ is a factor.

b Hence factor $f(x)$ completely. Dividing: $f(x) = (x + 3)(x^2 - 4) = (x + 3)(x - 2)(x + 2)$.

(a) $f(-3) = (-27) + 3(9) - 4(-3) - 12 = -27 + 27 + 12 - 12 = 0$, so by the factor theorem $(x + 3)$ is a factor.; (b) Dividing: $f(x) = (x + 3)(x^2 - 4) = (x + 3)(x - 2)(x + 2)$.

4 **[Foundational]** • **Target: Polynomial division, remainder & factor theorems**

Divide $x^3 - 2x^2 - 5x + 6$ by $(x - 3)$ using polynomial long division. State the quotient and remainder.

Quotient $= x^2 + x - 2$, remainder $= 0$. So $x^3 - 2x^2 - 5x + 6 = (x - 3)(x^2 + x - 2) = (x - 3)(x + 2)(x - 1)$.

5 **[Foundational]** • **Target: Exponential equations**

Solve $2^x = 15$ for x, giving your answer to three decimal places.

$x = \frac{\ln 15}{\ln 2} \approx \frac{2.708}{0.693} \approx 3.907$.

6 **[Foundational]** • **Target: Nonlinear systems of equations**

Solve the system $y = x^2 - 1$ and $y = 2x + 2$.

$x^2 - 1 = 2x + 2 \Rightarrow x^2 - 2x - 3 = 0 \Rightarrow (x - 3)(x + 1) = 0 \Rightarrow x = 3$ or $x = -1$. Solutions: $(3, 8)$ and $(-1, 0)$.

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7 [Foundational] • Target: Systems of inequalities and linear programming

A feasible region is bounded by the constraints $x \geq 0$, $y \geq 0$, $x + y \leq 6$, and $2x + y \leq 10$. Find the vertices of the feasible region.

Intersecting boundary lines pairwise (and checking each candidate point satisfies every constraint) gives vertices $(0, 0)$, $(5, 0)$, $(4, 2)$, and $(0, 6)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Systems of inequalities and linear programming

Using the feasible region from the previous question, find the maximum value of $P = 3x + 2y$ over the region, and state where it occurs.

Evaluating P at each vertex: $(0, 0) \to 0$; $(5, 0) \to 15$; $(4, 2) \to 16$; $(0, 6) \to 12$. The maximum is $P = 16$, occurring at $(4, 2)$.

9 [Intermediate] • Target: A well-designed word problem: production planning at a furniture workshop

This question has three parts. A furniture workshop makes tables and chairs. Each table requires 4 hours of carpentry and 2 hours of finishing; each chair requires 1 hour of carpentry and 1 hour of finishing. Each week the shop has at most 32 hours of carpentry time and 20 hours of finishing time available. Each table earns a profit of 70 dollars and each chair earns a profit of 30 dollars. Let x be the number of tables and y the number of chairs produced in a week. (a) Write the system of inequalities describing the constraints (including $x \geq 0$ and $y \geq 0$). (b) Find the vertices of the feasible region. (c) Determine the number of tables and chairs that maximizes the weekly profit $P = 70x + 30y$, and find the maximum profit.

(a) The constraints are $x \geq 0$, $y \geq 0$, $4x + y \leq 32$ (carpentry), and $2x + y \leq 20$ (finishing). (b) The boundary lines meet the axes and each other at $(0, 0)$, $(8, 0)$, $(6, 8)$, and $(0, 20)$ — these are the vertices of the feasible region. (c) Evaluating $P = 70x + 30y$: $(0, 0) \to 0$; $(8, 0) \to 560$; $(6, 8) \to 660$; $(0, 20) \to 600$. The maximum profit is 660 dollars, achieved by producing 6 tables and 8 chairs per week.

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10 [Intermediate] • Target: Simplifying rational expressions

Solve the nonlinear system of equations: $\{y = x^2 - 4x + 3; 2x + y = 6$ Find all points of intersection (x, y).

Step 1: Isolate y from the linear equation: $y = 6 - 2x$ Step 2: Substitute into the quadratic equation: $6 - 2x = x^2 - 4x + 3 \Rightarrow x^2 - 2x - 3 = 0$. Step 3: Factor the quadratic: $(x - 3)(x + 1) = 0 \Rightarrow x_1 = 3, \quad x_2 = -1$. Step 4: Compute corresponding y-values:For $x = 3$: $y = 6 - 2(3) = 0 \Rightarrow \mathbf{(3, 0)}$.For $x = -1$: $y = 6 - 2(-1) = 8 \Rightarrow \mathbf{(-1, 8)}$. The points of intersection are $\mathbf{(3, 0)}$ and $\mathbf{(-1, 8)}$.

11 [Intermediate] • Target: Absolute value equations

Find all real solutions to the rational equation $\frac{2}{x-3} - \frac{3}{x+1} = \frac{8}{x^2-2x-3}$.

Step 1: Factor denominators: $x^2 - 2x - 3 = (x - 3)(x + 1)$. LCD is $(x - 3)(x + 1)$. Restrictions: $xeq 3, -1$. Step 2: Multiply through by LCD: $2(x+1) - 3(x-3) = 8 \Rightarrow 2x + 2 - 3x + 9 = 8 \Rightarrow -x + 11 = 8 \Rightarrow -x = -3 \Rightarrow x = 3.$ Step 3: Check restriction: $x = 3$ is a restricted value causing division by zero in the original equation. Therefore, $x = 3$ is extraneous and the equation has no solution (empty set $\varnothing$).

12 [Intermediate] • Target: Polynomial division, remainder & factor theorems

When the polynomial $P(x) = 2x^3 - kx^2 + 5x - 7$ is divided by $(x - 2)$, the remainder is 11. Find the value of k.

Step 1: Apply the Remainder Theorem: The remainder when dividing by $(x - 2)$ is $P(2)$: $P(2) = 11$ Step 2: Evaluate $P(2)$: $P(2) = 2(2^3) - k(2^2) + 5(2) - 7 = 2(8) - 4k + 10 - 7 = 16 - 4k + 3 = 19 - 4k$ Step 3: Equate and solve for k: $19 - 4k = 11 \Rightarrow 4k = 19 - 11 = 8 \Rightarrow \mathbf{k = 2}$. The value of k is $\mathbf{2}$.

13 [Intermediate] • Target: Logarithmic equations

Solve the logarithmic equation $\log_2(x + 4) - \log_2(x) = 3$ for x.

Step 1: Apply quotient law of logarithms: $\log_2\left(\frac{x+4}{x}\right) = 3$ Step 2: Convert to exponential form: $\frac{x+4}{x} = 2^3 = 8$ Step 3: Solve for x: $x + 4 = 8x \Rightarrow 7x = 4 \Rightarrow \mathbf{x = \frac{4}{7}}$. Step 4: Check domain: For $x = 4/7$, both $x > 0$ and $x + 4 > 0$ are strictly positive ✓.

14 [Intermediate] • Target: Exponential equations

When the polynomial $P(x) = 2x^3 - kx^2 + 5x - 6$ is divided by $x - 2$, the remainder is 8. What is the value of the constant k?

By the Polynomial Remainder Theorem, the remainder upon division by $x - 2$ is equal to $P(2)$: $P(2) = 8$ $2(2^3) - k(2^2) + 5(2) - 6 = 8$ $2(8) - 4k + 10 - 6 = 8 \Rightarrow 16 - 4k + 4 = 8 \Rightarrow 20 - 4k = 8$ $-4k = -12 \Rightarrow k = \mathbf{3}$

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Nonlinear systems of equations** • **Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic**

Solve the exponential equation $3^{2x-1} = 5^{x+2}$ for x in terms of natural logarithms.

Step 1: Take the natural logarithm of both sides: $\ln(3^{2x-1}) = \ln(5^{x+2}) \Rightarrow (2x-1)\ln(3) = (x+2)\ln(5)$. Step 2: Expand and group all terms containing x: $2x\ln(3) - \ln(3) = x\ln(5) + 2\ln(5) \Rightarrow x[2\ln(3) - \ln(5)] = 2\ln(5) + \ln(3)$. Step 3: Solve for x: $x = \frac{\mathbf{2\ln(5) + \ln(3)}}{\mathbf{2\ln(3) - \ln(5)}} = \frac{\ln(\mathbf{75})}{\ln(\mathbf{9/5})} \approx 7.37$.

16 **[Challenging]** • **Target: Rational equations and extraneous solutions** • **Cross-Lesson: 3.01 Sequences, Series & Complex Numbers**

On the EST II Math exam, Advanced Algebra and Functions together account for 40% of the questions. In a 50-question test, if a student targets an accuracy rate of 90% in these two high-yield domains, how many questions from these domains can the student afford to miss at most?

Step 1: Find total questions from Advanced Algebra and Functions: $50 \times 0.40 = 20$ questions. Step 2: Calculate 90% accuracy target: $20 \times 0.90 = 18$ questions correct. Step 3: Find maximum allowed missed questions: $20 - 18 = 2$ questions.

17 **[Challenging]** • **Target: Systems of inequalities and linear programming** • **Cross-Lesson: 4.01 Lines, Triangles & Polygons**

Solve the rational equation for x: $\frac{x}{x-2} + \frac{1}{x+3} = \frac{5}{x^2+x-6}$. Check for extraneous solutions.

Factor the quadratic denominator: $x^2 + x - 6 = (x-2)(x+3)$. The LCD is $(x-2)(x+3)$, with restricted domain $x \neq 2, -3$. Multiply both sides by $(x-2)(x+3)$: $x(x+3) + 1(x-2) = 5$.
$x^2 + 3x + x - 2 = 5 \Rightarrow x^2 + 4x - 7 = 0$. Apply the quadratic formula:
$x = \frac{-4 \pm \sqrt{16 - 4(1)(-7)}}{2} = \frac{-4 \pm \sqrt{16+28}}{2} = \frac{-4 \pm \sqrt{44}}{2} = \frac{-4 \pm 2\sqrt{11}}{2} = -2 \pm \sqrt{11}$. Check restrictions: $-2 \pm \sqrt{11} \approx 1.317$ and -5.317. Neither equals.

18 **[Challenging]** • **Target: A well-designed word problem: production planning at a furniture workshop** • **Cross-Lesson: 5.01 Trigonometry**

Solve the logarithmic equation $\log_3(x+6) + \log_3(x) = 3$ and eliminate any extraneous solutions.

Combine the logarithms on the left-hand side using the product rule: $\log_3[x(x+6)] = 3$. Convert to exponential form: $x(x+6) = 3^3 = 27 \Rightarrow x^2 + 6x - 27 = 0$. Factor the quadratic:
$(x+9)(x-3) = 0 \Rightarrow x = -9$ or $x = 3$. Check the domain restrictions in the original equation:For $x = -9$: $\log_3(-9)$ is undefined, so $x = -9$ is an extraneous solution.For $x = 3$: $\log_3(9) + \log_3(3) = 2 + 1 = 3$. Valid ✓.Therefore, the unique solution is $x = 3$.

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19 [Challenging] • Target: Simplifying rational expressions • Cross-Lesson: 6.01 Circles & Coordinate Geometry

Find all solutions of $\cos(2\theta) = \cos\theta$ in the interval $0^\circ \le \theta \le 360^\circ$.

Step 1: Use double-angle identity: $\cos(2\theta) = 2\cos^2\theta - 1 \Rightarrow 2\cos^2\theta - 1 = \cos\theta \Rightarrow 2\cos^2\theta - \cos\theta - 1 = 0.$ Step 2: Factor the quadratic: $(2\cos\theta + 1)(\cos\theta - 1) = 0.$ Step 3: Solve factors:$\cos\theta = 1 \Rightarrow \mathbf{\theta = 0^\circ, 360^\circ}$.$\cos\theta = -\frac{1}{2} \Rightarrow$ Reference angle is 60°, in Quadrants II and III.

20 [Challenging] • Target: Absolute value equations • Cross-Lesson: 7.01 Solid Geometry & Similarity

Let $f(x) = \frac{2x+1}{x-3}$ for $x \neq 3$. (a) Determine the inverse function $f^{-1}(x)$. (b) State the domain and range of $f^{-1}(x)$.

(a) Find the inverse function: Set $y = \frac{2x+1}{x-3}$. Swap x and y:
$x = \frac{2y+1}{y-3} \Rightarrow x(y-3) = 2y+1 \Rightarrow xy - 3x = 2y + 1.$ Group y terms:
$xy - 2y = 3x + 1 \Rightarrow y(x-2) = 3x + 1 \Rightarrow \mathbf{f^{-1}(x) = \frac{3x+1}{x-2}}.$ (b) Domain and Range: The denominator of $f^{-1}(x)$ cannot be zero, so the domain is $\mathbf{(-\infty, 2) \cup (2, \infty)}$ (all $x \neq 2$). The range of f^{-1} equals the domain of f, which is.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Simplifying rational expressions

Solve the system $y = x^2 + 4x + 1$ and $y = -5x - 7$. Find the x-coordinates of the intersection points.

a. $-8, -3$

b. $-8, -1$ ✓

c. $-6, -1$

d. $8, 1$

$x^2 + 4x + 1 = -5x - 7 \Rightarrow x^2 + 9x + 8 = 0 \Rightarrow (x - (-8))(x - (-1)) = 0 \Rightarrow x = -8$ or $x = -1$.

22 [Foundational] • Target: Absolute value equations

Using the Factor Theorem, find the remainder when $f(x) = x^3 - x^2 - 6x - 4$ is divided by $(x + 2)$.

a. **-4** ✓

b. -9

c. 1

d. 5

Remainder $= f(-2) = -4$ (so it is *not* a factor).

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23 [Foundational] • Target: Polynomial division, remainder & factor theorems

In the system of linear equations below, k is a real constant. $(k-1)x+6y, =18; 2x+ky, =12$ For which value of k does the system have infinitely many solutions (x, y)?

a. -4

b. -3

c. 3

d. 4 ✓

For a system of two linear equations in two variables to have infinitely many solutions, the two equations must represent the exact same line. This requires the ratios of the corresponding coefficients and the constant terms to all be equal: $\frac{k-1}{2}=\frac{6}{k}=\frac{18}{12}$ First, simplify the ratio of the constant terms: $\frac{18}{12}=\frac{3}{2}$ Next, set the ratio of the y-coefficients equal to $\frac{3}{2}$: $\frac{6}{k}=\frac{3}{2}\Rightarrow 3k=12\Rightarrow k=4$ Verify this value.

24 [Foundational] • Target: Logarithmic equations

When $f(x)=x^2+3x+5$ is divided by $(x-3)$, what is the remainder?

a. 28

b. 26

c. 19

d. **23** ✓

By the Remainder Theorem, remainder $=f(3)=(3)^2+3\cdot(3)+5=23$.

25 [Foundational] • Target: Exponential equations

Consider the system of linear equations below: $\{3x-2y=8; kx+6y=5$ For which of the following values of the constant k does the system have no solution (x, y)?

a. **-9** ✓

b. -4

c. 4

d. 9

A system of two linear equations in two variables has no solution if and only if the lines represented by the equations are parallel and distinct (they have the same slope but different y-intercepts). First, rewrite each equation in slope-intercept form $y=mx+b$: 1. For the first equation:
$3x-2y=8\Rightarrow -2y=-3x+8\Rightarrow y=\frac{3}{2}x-4$ The slope of the first line is $m_1=\frac{3}{2}$ and the y-intercept is -4. 2. For the second equation.

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26 [Foundational] • Target: Nonlinear systems of equations

A quadratic function is given by $f(x) = x^2 - 4x + 3$. The graph of this function is a parabola. Which of the following points lies on the graph of $f(x)$?

a. $(1, 0)$ ✓

b. $(0, 1)$

c. $(2, 3)$

d. $(4, 4)$

To determine which point lies on the graph of $f(x) = x^2 - 4x + 3$, we substitute the x-coordinate of each point into the function and check if the resulting y-coordinate matches the y-coordinate of the point. For the point $(1, 0)$: Substitute $x = 1$ into $f(x)$: $f(1) = (1)^2 - 4(1) + 3 = 1 - 4 + 3 = 0$. Since $f(1) = 0$, the point $(1, 0)$ lies on the graph. For the point $(0, 1)$: Substitute $x = 0$ into $f(x)$: $f(0) = (0)^2 - 4(0) + 3 = 0 - 0 + 3 = 3$. Since $f(0) = 3eq1$, the point $(0, 1)$ does not lie.

27 [Foundational] • Target: Rational equations and extraneous solutions

In the standard (x, y) coordinate plane, the system of equations $\{y = |x^2 - 4|; y = 2x + k$ has exactly 3 distinct real solutions (x, y). What is the sum of all possible values of the real constant k?

a. 4

b. 5

c. **9** ✓

d. 11

The system represents the intersections of the absolute-value quadratic graph $y = |x^2 - 4|$ and the family of parallel lines $y = 2x + k$ of slope 2. The piecewise definition of $y = |x^2 - 4|$ is: \[y = \begin{cases} x^2 - 4, & \text{if } |x| \ge 2 \\ 4 - x^2, & \text{if } |x| 2 \) and $1 - \sqrt{10}5$: 2 solutions (outer branches only). - For $4 < k < 5$: 4 solutions (2 on the inverted arch, 2 on the outer branches). - For $-4 < k < 4$: 2 solutions. - For $k = -4$: 1 solution (passing through $(2, 0)$). - For.

28 [Foundational] • Target: Systems of inequalities and linear programming

Consider the function $h(x) = ax^2 + bx + c$. Given that $h(1) = 6$, $h(-1) = 10$, and the graph of $y = h(x)$ has a vertex at $(3, -10)$, find the value of $a + b + c$.

a. 12

b. **6** ✓

c. 10

d. 2

The function is given by $h(x) = ax^2 + bx + c$. We are given the following conditions: 1. $h(1) = 6$: $a(1)^2 + b(1) + c = 6$ $a + b + c = 6$ (Equation 1) 2. $h(-1) = 10$: $a(-1)^2 + b(-1) + c = 10$ $a - b + c = 10$ (Equation 2) 3. The vertex of the parabola $y = h(x)$ is at $(3, -10)$. The x-coordinate of the vertex of a parabola $y = ax^2 + bx + c$ is given by $x_v = -\frac{b}{2a}$. So, $3 = -\frac{b}{2a}$ which implies $b = -6a$ (Equation 3). The y-coordinate of the vertex is $h(3) = -10$..

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: A well-designed word problem: production planning at a furniture workshop**

Consider the function $g(x) = ax^2 + bx + c$. The graph of $y = g(x)$ is a parabola that passes through the points $(1, 5)$, $(2, 12)$, and $(3, 21)$. If the vertex of the parabola is at (h, k), what is the value of h?

a. $\frac{5}{2}$ ✓

b. 2

c. $-\frac{5}{2}$

d. -2

Let the quadratic function be $g(x) = ax^2 + bx + c$. We are given three points that lie on the graph of $y = g(x)$: 1. $(1, 5) \Rightarrow a(1)^2 + b(1) + c = 5 \Rightarrow a + b + c = 5$ (Equation 1) 2. $(2, 12) \Rightarrow a(2)^2 + b(2) + c = 12 \Rightarrow 4a + 2b + c = 12$ (Equation 2) 3. $(3, 21) \Rightarrow a(3)^2 + b(3) + c = 21 \Rightarrow 9a + 3b + c = 21$ (Equation 3) Subtract Equation 1 from Equation 2: $(4a + 2b + c) - (a + b + c) = 12 - 5 \Rightarrow 3a + b = 7$ (Equation 4) Subtra.

30 **[Intermediate]** • **Target: Simplifying rational expressions**

Consider the function $f(x) = ax^2 + bx + c$. The graph of $y = f(x)$ passes through the points $(1, 2)$, $(2, 7)$, and $(3, 14)$. Find the value of a.

a. **1** ✓

b. 2

c. 3

d. 4

We are given that the quadratic function $f(x) = ax^2 + bx + c$ passes through the points $(1, 2)$, $(2, 7)$, and $(3, 14)$. We can set up a system of linear equations by substituting these points into the function: For $(1, 2)$: $a(1)^2 + b(1) + c = 2$ => $a + b + c = 2$ (Equation 1) For $(2, 7)$: $a(2)^2 + b(2) + c = 7$ => $4a + 2b + c = 7$ (Equation 2) For $(3, 14)$: $a(3)^2 + b(3) + c = 14$ => $9a + 3b + c = 14$ (Equation 3) Subtract Equation 1 from Equation 2: $(4a + 2b + c) - (a + b + c) = 7 - 2$ $3a + b = 5$.

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31 [Intermediate] • Target: Absolute value equations

Consider the function $g(x) = x^2 - ax + b$. If the graph of $y = g(x)$ has a minimum value of -1 at $x = 3$, find the values of a and b.

a. $a = 6, b = 8$ ✓

b. $a = -6, b = 8$

c. $a = 6, b = -1$

d. $a = -6, b = -1$

The function is a quadratic $g(x) = x^2 - ax + b$. The x-coordinate of the vertex (where the minimum occurs) is given by $-(-a)/(2 \cdot 1) = a/2$. We are given that the minimum occurs at $x = 3$. Therefore, $a/2 = 3$, which implies $a = 6$. The minimum value of the function is $g(3)$. Substituting $a = 6$ into the function, we get $g(x) = x^2 - 6x + b$. The minimum value is $g(3) = 3^2 - 6(3) + b = 9 - 18 + b = -9 + b$. We are given that the minimum value is -1. So, $-9 + b = -1$, which implies $b = 8$. Thus.

32 [Intermediate] • Target: Polynomial division, remainder & factor theorems

If $f(x) = x^2 + 1$ and $g(x) = -x - 1$, find the sum of all solutions to $(f \circ g)(x) = 5$.

a. **-2** ✓

b. 4

c. -6

d. -10

$(-x-1)^2 + 1 = 5 \Rightarrow (-x-1)^2 = 4 \Rightarrow -x - 1 = \pm 2 \Rightarrow x = -3$ or 1; sum $= -2$.

33 [Intermediate] • Target: Logarithmic equations

When $f(x) = x^4 + x^3 + 5x^2 - 5x - 1$ is divided by $(x + 3)$, the quotient is a cubic $q(x)$. What is the sum of the coefficients of $q(x)$?

a. -37

b. -33

c. -23

d. **-28** ✓

Synthetic division gives $q(x) = x^3 - 2x^2 + 11x - 38$, remainder 113. Sum of q's coefficients $= q(1) = -28$.

34 [Intermediate] • Target: Exponential equations

If y varies directly as x and $y = 12$ when $x = 3$, find y when $x = 7$.

a. 21

b. 24

c. **28** ✓

d. 25

Applying Advanced Algebra principles, $k = 4$; $y = 4(7) = 28$ Therefore, the verified correct option is 28.

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35 [Intermediate] • Target: Nonlinear systems of equations

For $x > 0$, solve $x + \frac{10}{x} = 7$ and give the larger root.

a. 9

b. **5 ✓**

c. -2

d. 11

Multiply by x: $x^2 - 7x + 10 = 0 \Rightarrow (x-2)(x-5) = 0 \Rightarrow x = 2, 5$; the larger is 5.

36 [Intermediate] • Target: Rational equations and extraneous solutions

Consider the function $g(x) = 2x + 3$. What is the value of $g(4)$?

a. 7

b. 8

c. **11 ✓**

d. 14

To find the value of $g(4)$, we substitute $x = 4$ into the function $g(x) = 2x + 3$. \\ x is 11.

37 [Intermediate] • Target: Systems of inequalities and linear programming

Divide $f(x) = x^3 + 3x^2 + 2x - 8$ by $(x - 3)$. What is the constant term of the quotient?

a. 18

b. 27

c. 25

d. **20 ✓**

Synthetic division by 3 gives quotient $x^2 + 6x + 20$ with remainder 52; constant term $= 20$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A well-designed word problem: production planning at a furniture workshop • Cross-Lesson: 5.01 Trigonometry

Two similar solids have a linear scale factor of $2:1$. If the surface area of the smaller solid is 12, find the surface area of the larger solid.

a. 14

b. 2

c. 13

d. **3 ✓**

Surface-area ratio $= \left(\frac{1}{2}\right)^2 \Rightarrow SA_2 = 12 \times \frac{1}{4} = 3$.

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39 [Challenging] • Target: Simplifying rational expressions • Cross-Lesson: 6.01 Circles & Coordinate Geometry

Two similar solids have a linear scale factor of $1:4$. If the volume of the smaller solid is 13, find the volume of the larger solid.

a. 443

b. 655

c. **832** ✓

d. 193

Volume ratio $=\left(\frac{4}{1}\right)^3 \Rightarrow V_2 = 13 \times \frac{64}{1} = 832$.

40 [Challenging] • Target: Absolute value equations • Cross-Lesson: 7.01 Solid Geometry & Similarity

What is the larger solution to $|x-5|=3$?

a. 10

b. **8** ✓

c. 2

d. 14

$x-5=\pm 3 \Rightarrow x=5+3=8$ or $x=5-3=2$; the larger is 8.

41 [Challenging] • Target: Polynomial division, remainder & factor theorems • Cross-Lesson: 8.01 Statistics & Probability

The line $y=-12$ intersects the circle $x^2+y^2=169$ at two points. What is the distance between the two intersection points?

a. **10** ✓

b. 13

c. 16

d. 5

$x^2=169-(-12)^2=25 \Rightarrow x=\pm 5$. Distance $=2(5)=10$.

42 [Challenging] • Target: Logarithmic equations • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

Simplify $\frac{x^2-16}{x-4}$, then evaluate the result at $x=8$.

a. 6

b. 14

c. 11

d. **12** ✓

$\frac{x^2-16}{x-4} = x+4$ (for $x \neq 4$), so at $x=8$: $8+4=12$.

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43 [Challenging] • Target: Exponential equations • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

The line $y = 4$ meets the circle $x^2 + y^2 = 25$ at a point with positive x-coordinate. Find that x-coordinate.

a. -2

b. 7

c. 11

d. **3 ✓**

$x^2 = 25 - 16 = 9 \Rightarrow x = 3$ (taking the positive root).

44 [Challenging] • Target: Nonlinear systems of equations • Cross-Lesson: 4.01 Lines, Triangles & Polygons

Two similar solids have a volume ratio of $125 : 1$. Find the linear (side-length) scale factor.

a. 1:9

b. 1:10

c. **1:5 ✓**

d. 1:7

Linear ratio $= \sqrt[3]{125 : 1} = 1 : 5$, since $1^3 : 5^3 = 125 : 1$.

45 [Challenging] • Target: Rational equations and extraneous solutions • Cross-Lesson: 5.01 Trigonometry

The line $y = -2x + 2$ intersects the parabola $y = x^2 + x + 4$ at two points. What is the sum of the x-coordinates of the intersection points?

a. **-3 ✓**

b. 1

c. -12

d. 3

Setting equal: $x^2 + 3x + 2 = 0$. Sum of roots $= -(3) = -3$.

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